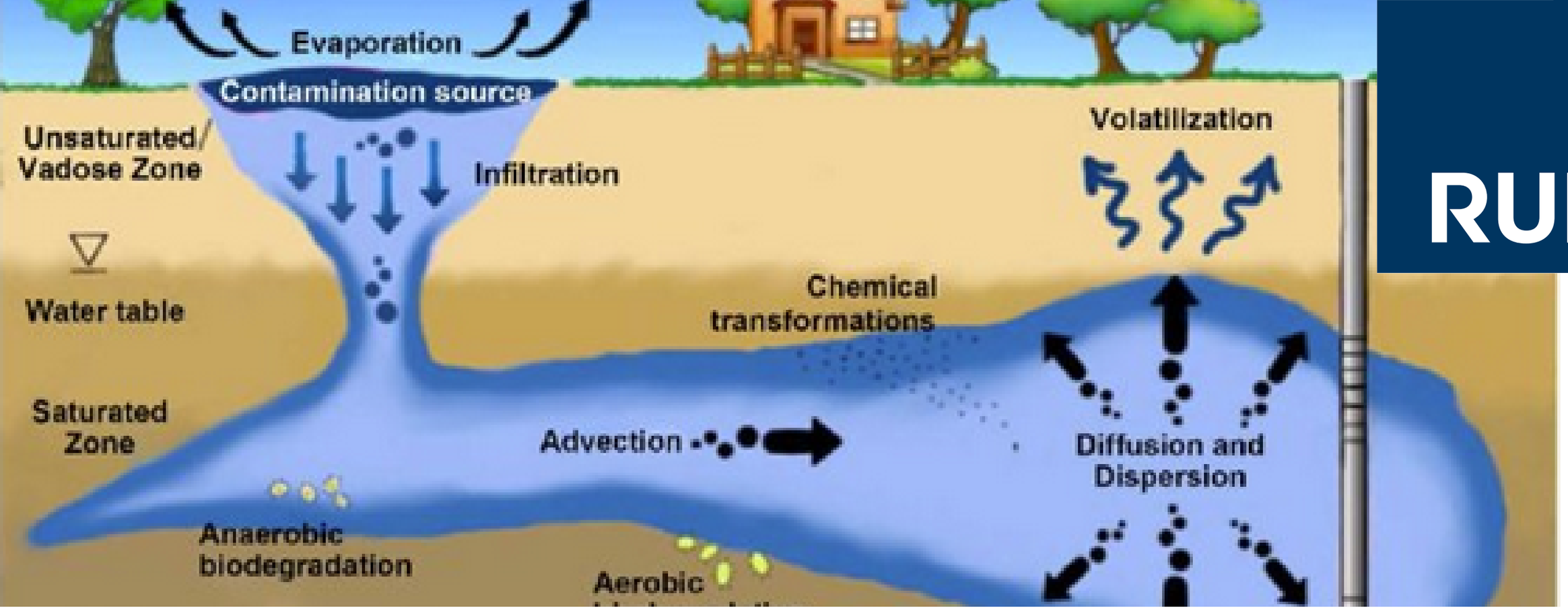


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GLOBAL CHANGE HYDROLOGY (GCH)

Processes in Water Quality

Prof. Martina Flörke

Chair of Engineering Hydrology and Water Resources Management

Classification of Emission Sources

Point sources: Discharge of pollutants into water bodies at a specific point

- **Indirect discharge:** Wastewater is discharged into the sewerage system and thus only indirectly into the receiving water body, i.e. usually via municipal sewage treatment plants.
- **Direct discharge:** Wastewater is discharged directly into the receiving water body



Diffuse sources: Pollutants are introduced over a large, undefined area

- Agricultural land
- Atmospheric deposition



Diffuse Sources: Runoff from Land

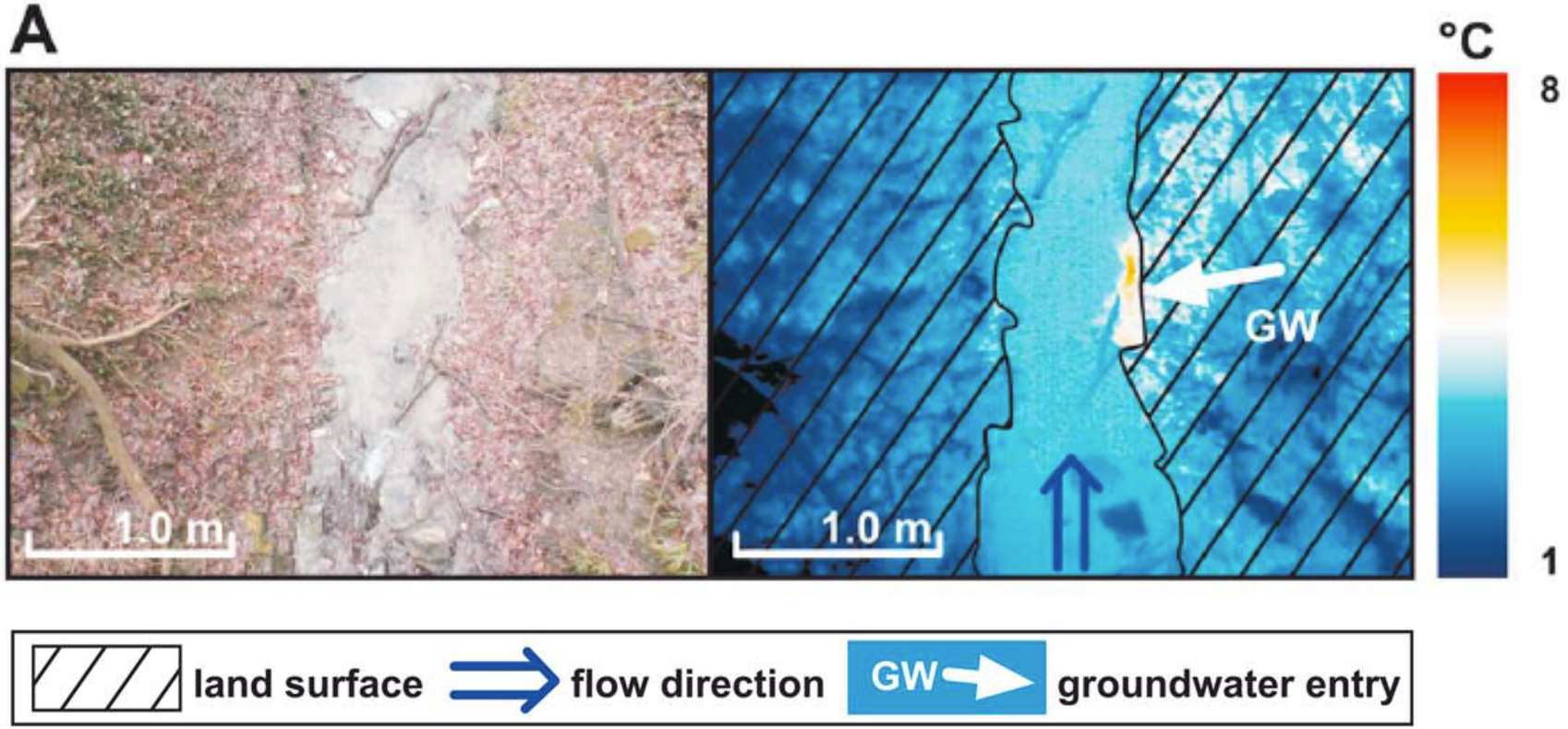
- **Direct discharge into water bodies along a river course**
- **Discharge of sediment-bound substances through erosion and sediment transport**
- **Indirect discharge via drainage systems after passing through the soil**





Oberflächenabfluss von einem Feld in den Vorfluter (Foto: Ingenieurbüro Feldwisch)

Diffuse Source: Groundwater



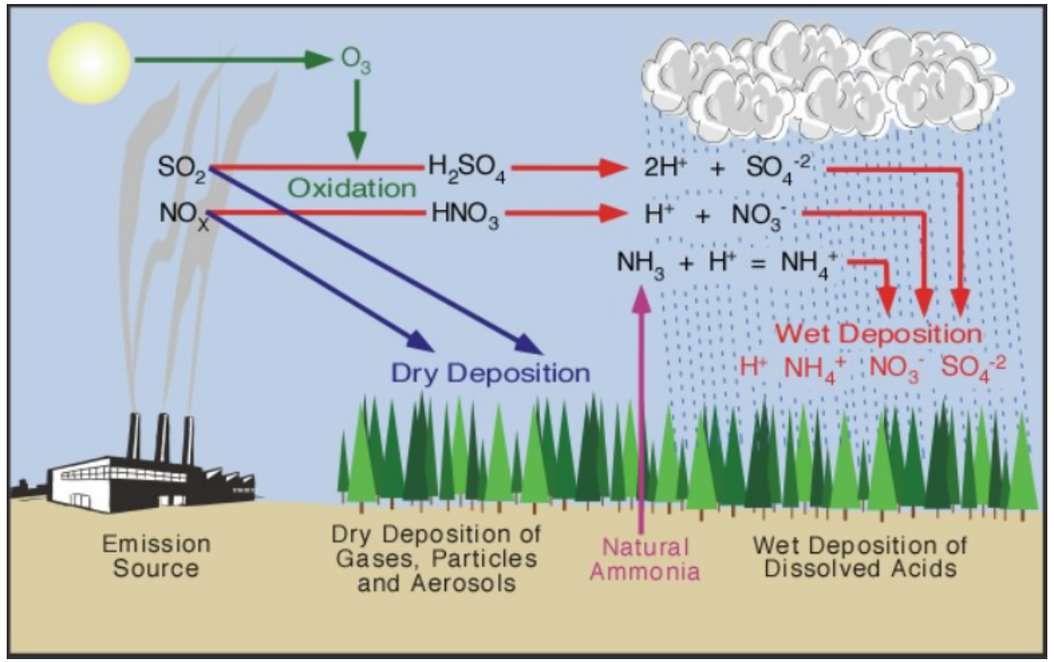
- Identification of a groundwater inflow using a thermal imaging camera

Schuetz & Weiler (2011)

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Diffuse Source: Atmospheric Deposition

- **Widespread deposition of gaseous or particulate matter suspended in the atmosphere**
- **Wet atmospheric deposition** (transfers air pollutants and nutrients directly into aquatic ecosystems through rain, snow, and fog)
- **Dry atmospheric deposition** (transfers air pollutants, dust, gases, and nutrients directly onto land and water surfaces without the aid of precipitation)



Quelle: S. Banzhaf

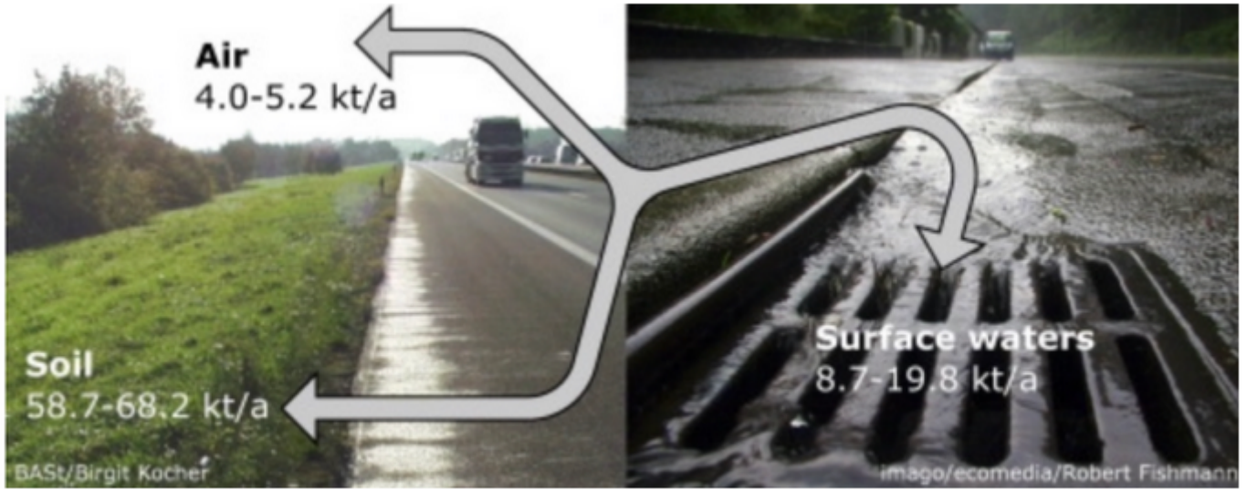
Diffuse Sources: Road Runoff

- **Deposits of pollutants on roads**

→ **Deposits of pollutants on roads**

Pollutants:

- Combustion residues
- Petrol additives
- Oil
- Heavy metals
- Microplastics (tyre wear)
- Salt
- …



Wo der Reifenabrieb landet (Q. BASt)

Conservative Pollutant Transport

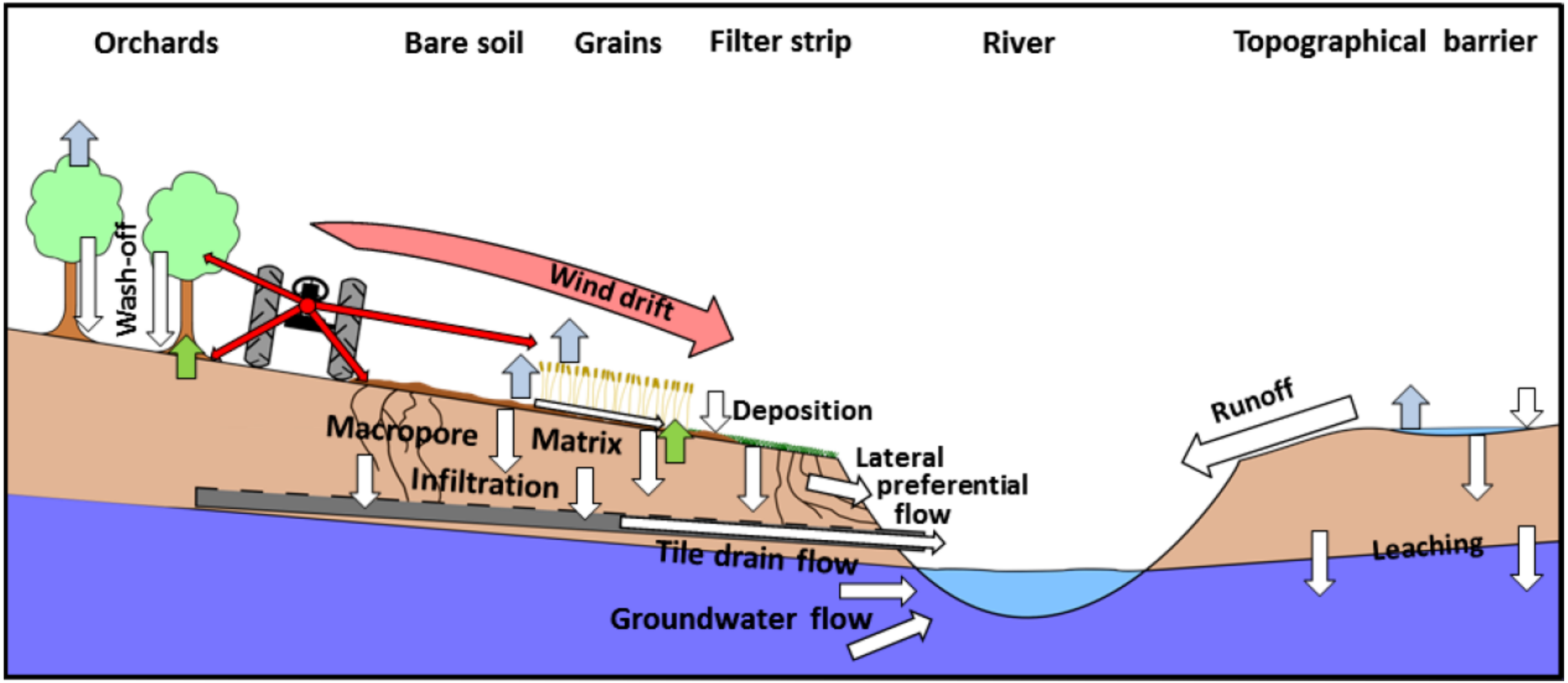
Definition:

Conservative pollutant transport describes the movement of a substance through water or soil without undergoing any chemical decay, biological transformation, or loss of mass.

Key concepts:

1. **Passive Movement:** The pollutant travels with the fluid like a passive tracer (such as salt).
2. **No Decay:** The pollutant does not break down over time or react with surrounding sediments or organic matter.
3. **Mass Conservation:** The total mass of the pollutant remains constant throughout the transport processes.
4. **Main Transport Mechanisms:** These include advection (convection), dispersion and diffusion.

Transport Processes



Gaßmann (2013)

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General Transport Equation

transport terms | reactive terms

$$\frac{\partial C_i}{\partial t} = D_l \frac{\partial^2 C_i}{\partial z^2} + D_t \frac{\partial^2 C_i}{\partial z^2} + D \frac{\partial^2 C_i}{\partial z^2} - v \frac{C_i}{\partial z} + C_{QS}$$

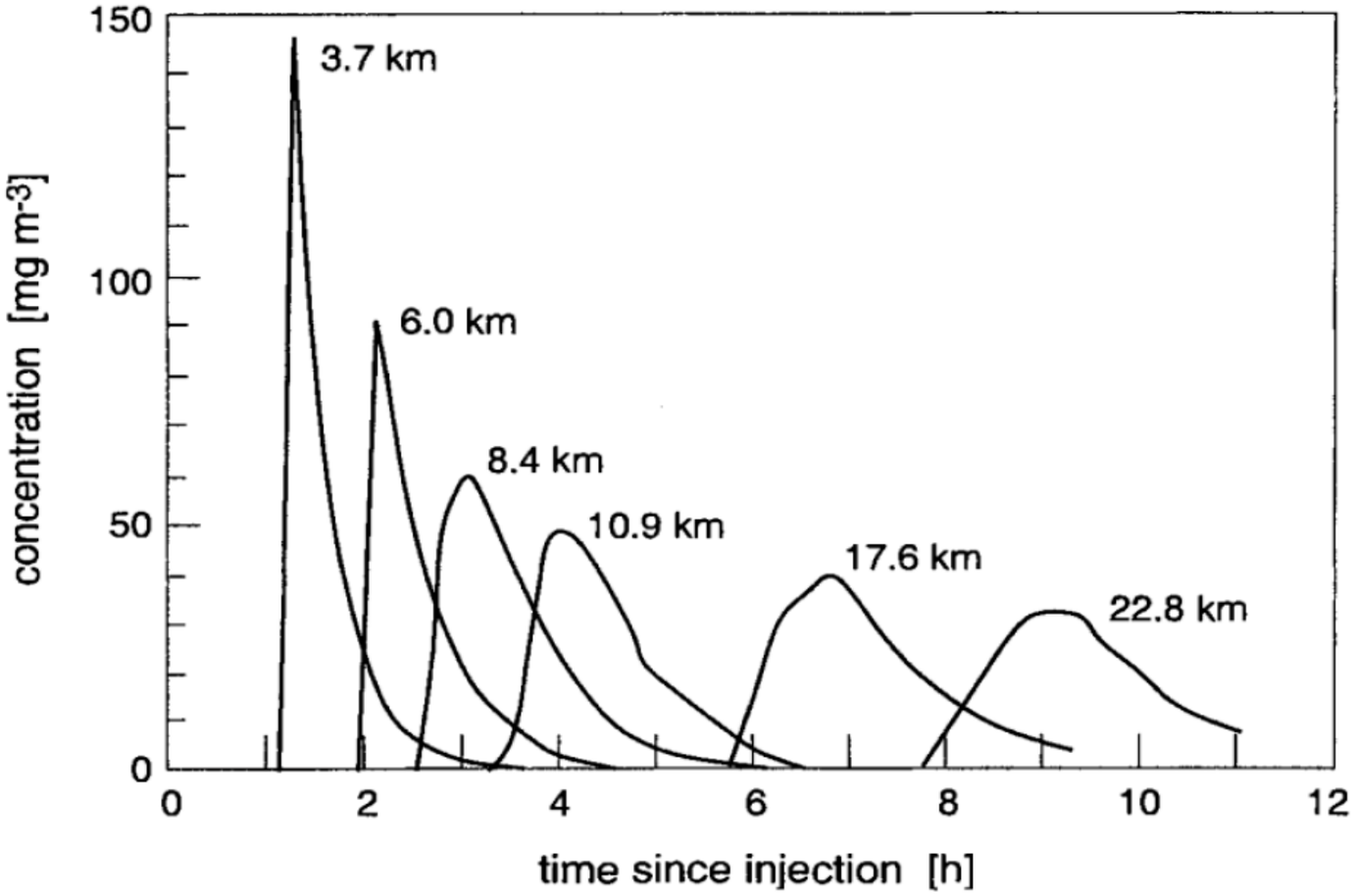
Dispersion | Diffusion | Advection | Source/Sink

C_i	= concentration of species dissolved in water i [mol/L]
t	= time [s]
D_l	= longitudinal dispersion coefficient [m²/s]
D_t	= transversal dispersion coefficient [m²/s]
D	= diffusion coefficient [m²/s]
z	= spatial coordinate [m]
v	= distance velocity [m/s]
C_{QS}	= concentration of substance (source or sink)

Merkel & Planer-Friedrich (2002)

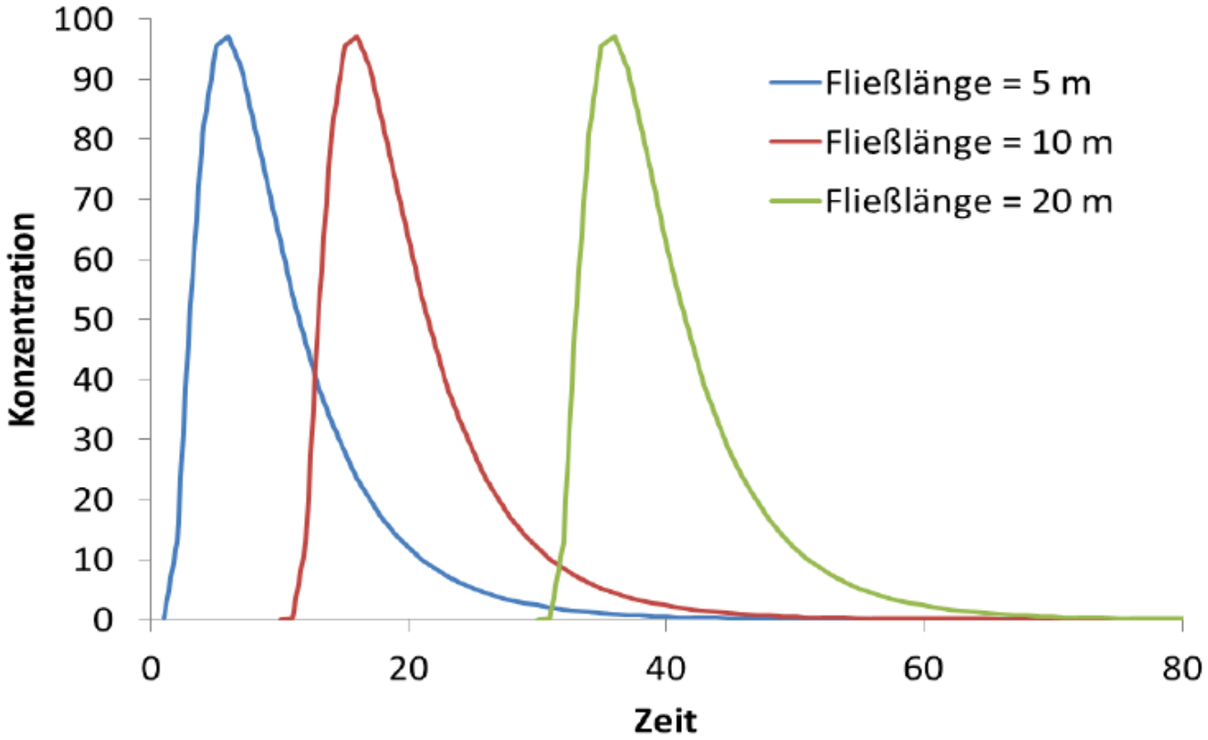
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Trend of a Concentration Curve in a River (or Stream)



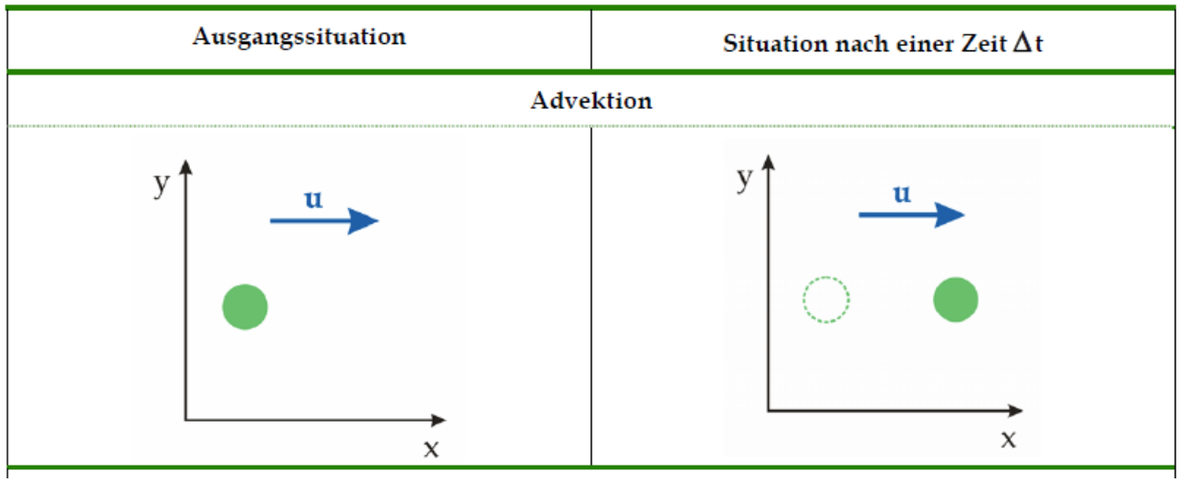
Schwarzenbach et al. (2003)

Advection



Quelle: M. Gaßmann

- Transport by flow of water



Kühlers et al. (2010)

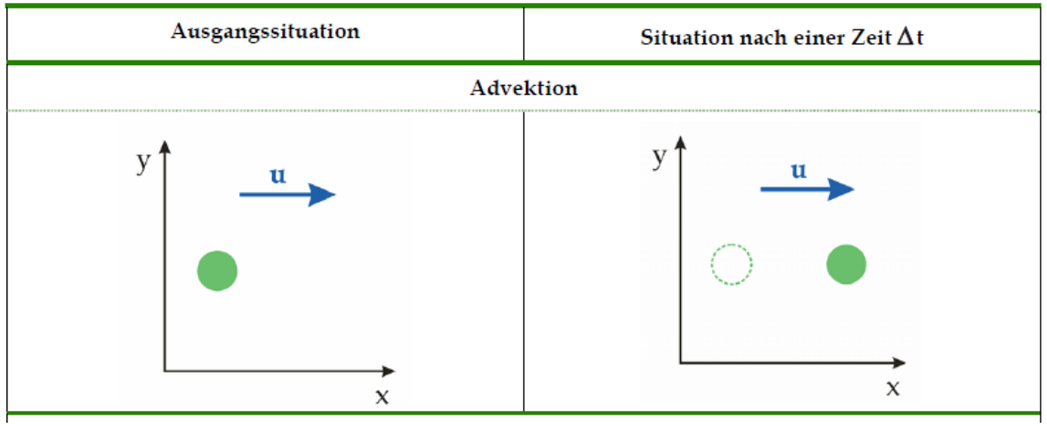
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Advective Transport

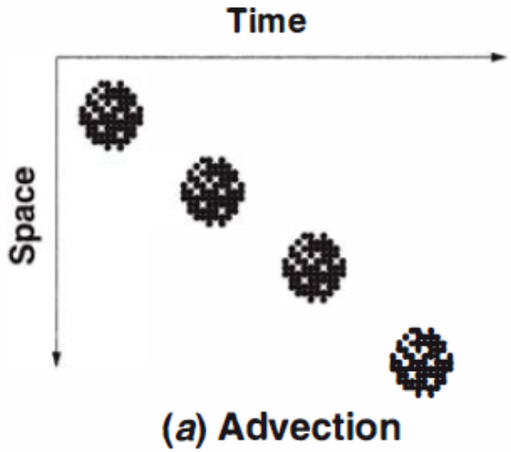
Due to advection at a velocity u, the colour spot is displaced by a distance u·Δt over a time interval Δt, without changing its shape, intensity or concentration. This process generally results in the predominant transport of a substance in the direction of flow.

The advective transport of a substance results in a specific mass flow rate j [kg/(m²·s)] (displacement of a mass of substance per unit time through a control surface), which can be calculated as the product of the concentration C [kg/m³] of the substance and the specific flow rate q [m/s] of the underlying flow.

$$j = q \cdot C$$



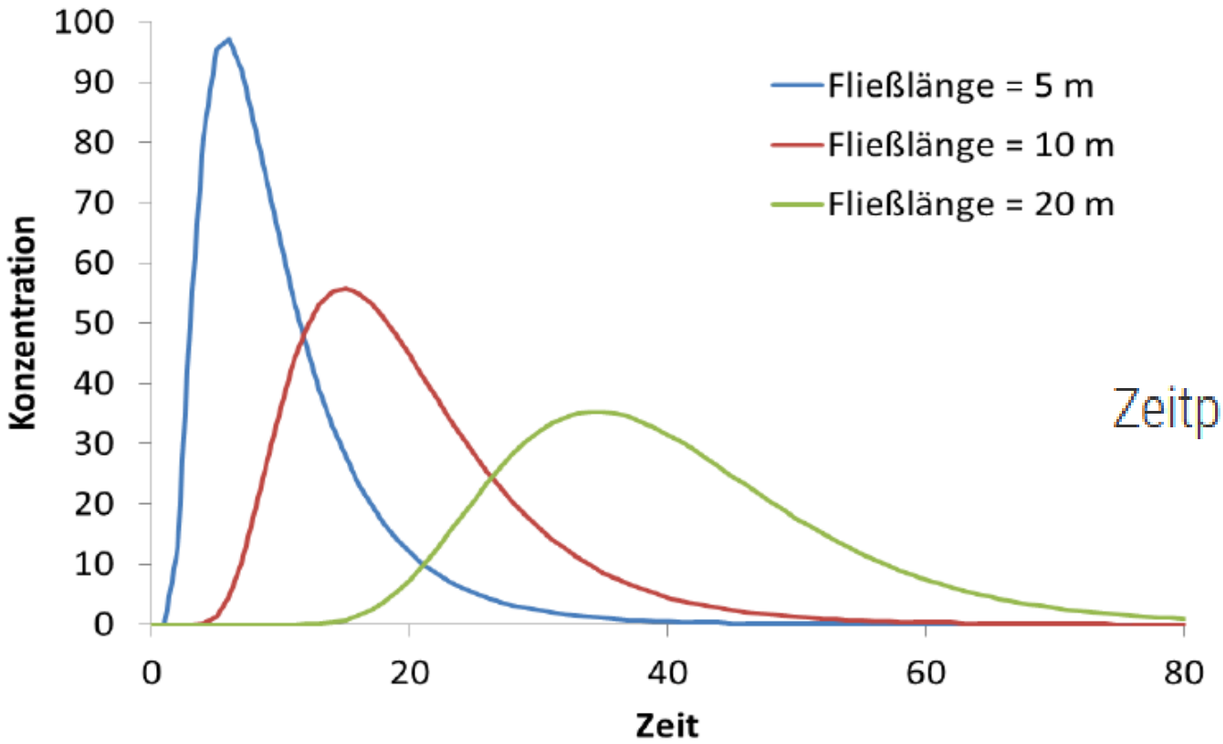
Kühlers et al. (2010)



Chapra (2008)

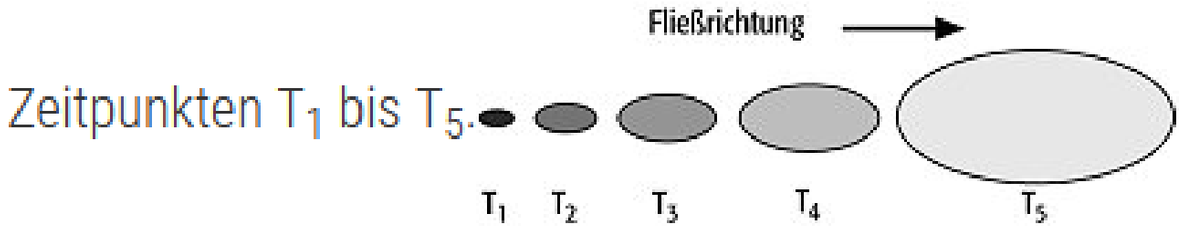
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Dispersion



Quelle: M. Gaßmann

- Transport by flow of water



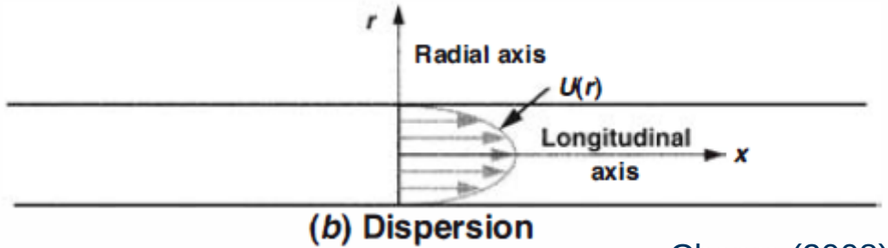
www.spektrum.de

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Dispersion Transport

The spatial variation in the time-averaged flow velocity (e.g. high flow velocities in the centre of a channel or a cavity in a porous medium, low flow velocities at the edges) causes additional mixing, known as dispersion.

A distinction is made between **longitudinal** dispersion, which occurs in the direction of flow (x-direction), and **transverse** dispersion, which occurs perpendicular to the direction of flow (y-direction).



Chapra (2008)

- As a result of longitudinal dispersion, for example, a concentration peak is smoothed out downstream; as lateral dispersion, it causes the substance to mix and spread from a point source of discharge across the width of the river.

Maniak (2010)

Dispersion coefficients in Rivers

Estimates according to a) Fischer et.al. (1979) or b) McQuivey and Keefer (1974)

a)

$$D = 0{,}011 \cdot \frac{U^2 \cdot B^2}{H \cdot U^*}$$

mit $U^* = \sqrt{g \cdot H \cdot S}$

b)

$$D = 0{,}058 \cdot \frac{Q}{S \cdot B}$$

For Froude number < 0,5

with $F = \frac{U}{\sqrt{g \cdot H}}$

U = flow velocity [m/s]
Q = average discharge [m^3/s]
B = river width [m]
H = average river depth [m]
U* = shear stress rate [m/s]
g = gravity constant [m/s^2]
S = river slope [-]

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Dispersion coefficients in Rivers

Examples: Hydrogeometric Parameters of Rivers, listed by river discharge (Fischer et al. 1979, in Chapra 2008)

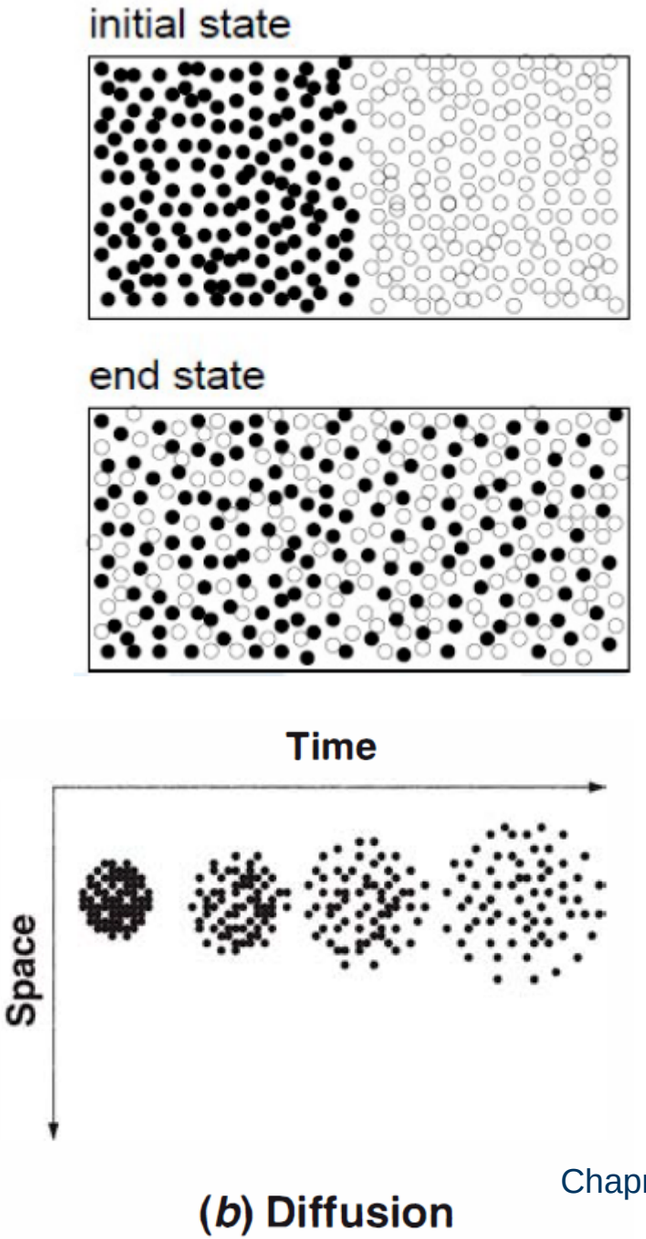
River	Mean depth (m)	Width (m)	Slope	Velocity (mps)	Flow ($m^3\ s^{-1}$)	Dispersion (0,1 $m^2\ s^{-1}$)
Missouri	2.70	200	0.00021	1.55	837.0	150.00
Sabine	3.40	116	0.00013	0.61	254.6	49.30
Windy/Big Horn	1.63	64	0.00135	1.22	144.1	10.10
Yadkin	3.10	71	0.00044	0.60	140.1	18.50
Clinch, Tennessee	1.68	53	0.00054	0.70	74.5	3.83
John Day	1.53	30	0.00239	0.92	41.8	3.95
Nooksack	0.76	64	0.00979	0.67	32.6	3.50
Coachella Canal, California	1.56	24	—	0.71	26.6	0.96
Bayou Anacoco	0.93	32	0.00050	0.37	10.9	3.60
Cinch, Virginia	0.58	36	—	0.21	4.4	0.81
Powell, Tennessee	0.85	34	—	0.15	4.3	0.95
Copper, Virginia	0.56	17	0.00130	0.32	3.6	1.51
Comite	0.43	16	0.00059	0.37	2.5	1.40

Diffusion

Diffusion is responsible for the mixing of the water. This mixing is caused by various mechanisms:

Molecular diffusion, caused by the intrinsic thermal motion of molecules, results in mass transport without an external energy input when a concentration gradient is present. The coloured spot gradually enlarges, the edges become blurred, and the maximum concentration decreases. Molecular diffusion acts isotropically, i.e. it is of equal magnitude in every spatial direction. It is a comparatively slow process.

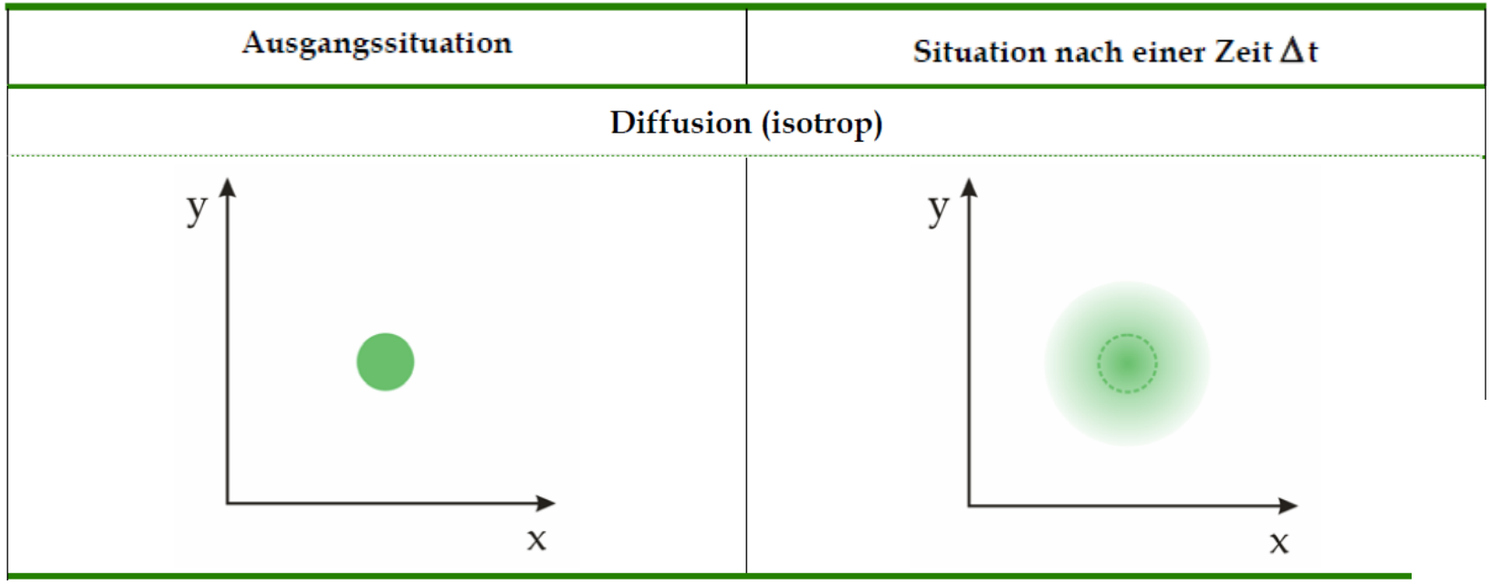
Turbulent diffusion is caused by the turbulent whirls of a flow. Mass transport is determined here not by the properties of the substance, but by the characteristics of the flow. It is therefore anisotropic. In flowing watercourses, turbulent diffusion causes much more intense mixing than molecular diffusion. In porous media, no turbulent diffusion takes place due to laminar flow.



Chapra (2008)

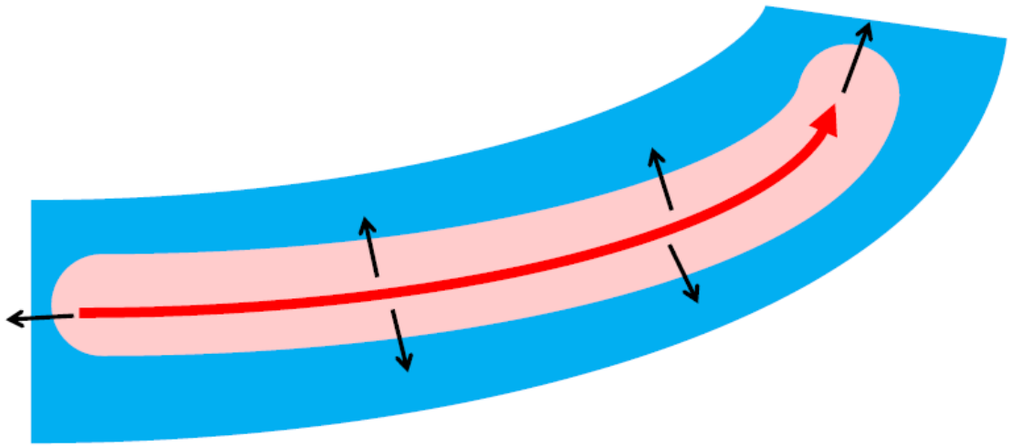
Diffusion

- **Levelling of concentration differences through Brownian motion**
- **Only relevant at very low flow rates (groundwater)**

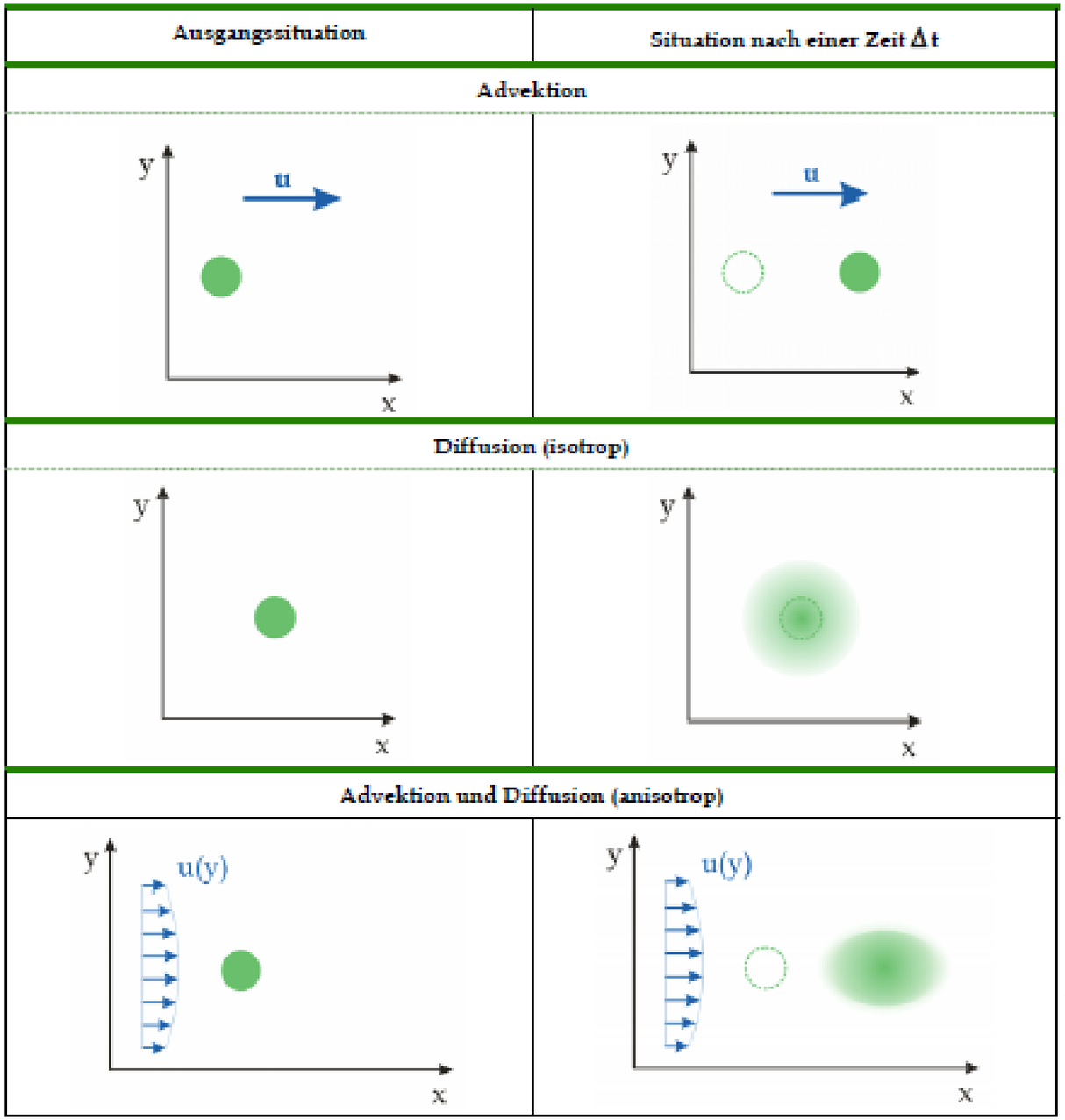


Kühlers et al. (2010)

➢ Transport by flow of water



Transport mechanisms



Kühlers et al. (2010)

Dilution

A constant quantity of a substance is introduced (e.g. via wastewater discharge from an industrial plant); the increase in water flow (caused by rainfall) leads to dilution in the water body and thus to falling concentrations.

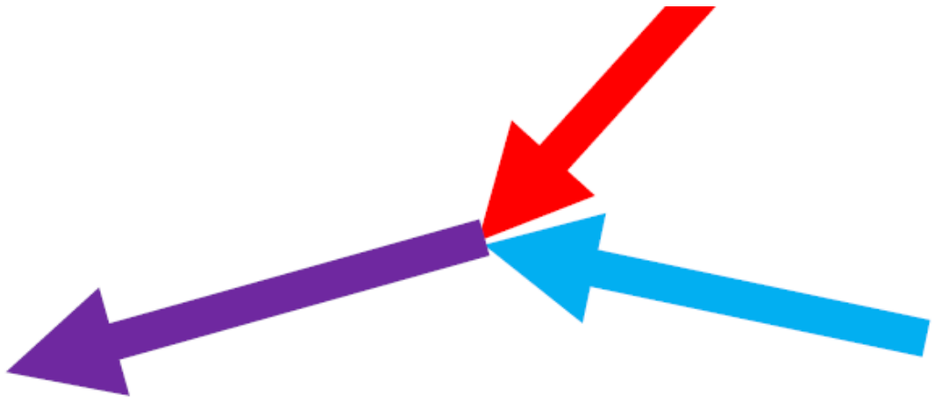
Mixing of two water bodies with different concentrations:

$$Mass\ conservation: M = M_1 + M_2$$

$$M = C \cdot Q$$

$$==> \quad C \cdot Q = C_1 \cdot Q_1 + C_2 \cdot Q_2$$

$$Q = Q_1 + Q_2$$



M = mass flow [g/s]
C = concentration [g/l]
Q = river discharge [l(s]

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Take Home Messages

- **Transport processes cause pollutants to move from one place to another**
- **The mass of the substances remains constant**
- **Peak concentrations and concentration trends are influenced by transport processes**
- **Key processes involved in conservative mass transport are:**
  - Advection
  - Diffusion
  - Dispersion
  - Diffusion and dispersion are often grouped together = hydrodynamic dispersion
  - In river basins/rivers, dispersion usually >> diffusion (except in groundwater)